

Seasonal Adjustment of the IIP Index

A Comparative Analysis of Time Series Methods

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Abstract

The Index of Industrial Production (IIP) is a key high-frequency barometer of industrial activity, but its raw monthly movements are dominated by recurring seasonal and calendar effects that obscure the underlying signal. This paper conducts a comprehensive seasonal adjustment (SA) of the IIP for the State of Odisha (base year 2011–12=100, monthly, April 2012–March 2023, $n=132$) and compares three families of methods: (i) classical multiplicative decomposition; (ii) the X-13ARIMA-SEATS program of the U.S. Census Bureau, in four operational variants; and (iii) a SARIMA-based approach. The General Index shows strong, broadly stable seasonality: production peaks at the fiscal year-end (March, +11.7% of the mean) and in winter, and troughs through the south-west monsoon (September, –9.0%). Adjustment compresses the index range from 197.3 to 165.7 points and lowers the coefficient of variation from 25.2% to 23.9%. All four X-13ARIMA-SEATS variants yield near-identical adjusted series (pairwise correlations 0.991–0.998), indicating robustness to the choice of filter; the X-13ARIMA-SEATS-labelled variant attains the lowest residual AICc (1109.6), and all variants leave residuals indistinguishable from white noise. Adjustment also delivers parsimony: the AICc-optimal raw-series model is ARIMA(1, 1, 4) with drift, whereas the adjusted series reduces to ARIMA(0, 1, 1) with drift. On the diagnostic evidence and its consistency with the practice of India’s National Statistics Office, we recommend the X-13ARIMA-SEATS framework for the routine seasonal adjustment of Odisha’s IIP. We also document the COVID-19 shock of 2020 and several item-level data-quality issues that condition interpretation.

Keywords: Index of Industrial Production; Seasonal adjustment; X-13ARIMA-SEATS; SARIMA; Classical decomposition; Odisha.

JEL Classification: C22; C43; C82; E32; R11.

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1 Introduction

The Index of Industrial Production (IIP) measures the relative change in the volume of industrial output over time with respect to a fixed base period. As a monthly indicator available with a relatively short lag, it is among the most closely watched short-term economic statistics in India: it feeds into the advance and provisional estimates of Gross Domestic Product, informs monetary-policy deliberations of the Reserve Bank of India, and is used by central and state governments to monitor the health of the industrial sector ([Reserve Bank of India, 2023](#)). At the sub-national level, a state IIP plays the same role for regional economic management, helping administrators distinguish genuine turning points in industrial activity from transient fluctuations.

A well-known difficulty, however, is that the raw IIP is contaminated by movements that are predictable and non-economic in origin. Industrial output responds systematically to the composition of the calendar (the number of working days in a month, the placement of week-ends, leap years), to the climate (in India, most prominently the south-west monsoon), and to festivals and holidays whose timing recurs each year. These *seasonal* and *calendar* effects can be large enough to dominate the month-to-month change in the index, so that a mechanical comparison of, say, March with the following July may suggest a sharp contraction when in fact nothing has changed in the underlying pace of production ([Ministry of Statistics and Programme Implementation, National Statistics Office, 2023](#)). Seasonal adjustment—also called deseasonalisation—is the statistical procedure that removes these predictable components so that the trend-cycle and genuinely new (irregular) developments become visible.

This paper studies the seasonal adjustment of the IIP for the State of Odisha. Odisha is an instructive case. Its industrial structure is unusually concentrated in mining and metals: in the index studied here, Mining & Quarrying and Manufacturing carry almost equal weight, and within manufacturing the single division of *Basic Metals* accounts for roughly one-third of the entire General Index. Both mining and metals are sensitive to the monsoon and to the fiscal-year production cycle, so one would expect Odisha’s IIP to display pronounced and economically interpretable seasonality. Establishing whether this is so, quantifying the seasonal pattern, and determining which adjustment method is most suitable are the practical questions that motivate the study.

The contribution of the paper is threefold. First, to our knowledge it provides the first systematic, multi-method seasonal adjustment of the *state-level* IIP for Odisha, covering the General Index and all of its constituent National Industrial Classification (NIC-2008) series. Second, rather than committing to a single technique, it places three families of methods side by side—classical decomposition, the X-13ARIMA-SEATS family, and a SARIMA-based approach—and compares them on a common footing using formal diagnostics (residual whiteness, information criteria, and the stability and dispersion of the adjusted series). Third, it translates the statistical results into the language of policy: it shows concretely *what is gained* from seasonal adjustment for Odisha, echoing the question posed for the all-India index by [Pandey et al. \(2020\)](#), and it issues a reasoned recommendation for the method that the state statistical machinery could adopt routinely.

The remainder of the paper is organised as follows. Section 2 reviews the evolution of seasonal-adjustment methodology and the Indian practice. Section 3 describes the data. Section 4 sets out the three methods and the diagnostic framework. Section 5 presents the empirical results method by method. Section 6 compares the methods and identifies the most suitable one. Section 7 interprets the seasonal pattern economically and discusses the COVID-19 episode and data quality. Section 8 concludes with policy implications, limitations, and directions for future work. An appendix reports the full adjusted series and supplementary diagnostics.

2 Review of Literature

2.1 The evolution of seasonal-adjustment methods

The decomposition of an economic time series into trend-cycle, seasonal, and irregular components has a long history. The dominant operational tradition descends from the moving-average “ratio-to-moving-average” methods developed at the U.S. Bureau of the Census, codified in the *X-11* variant of Census Method II (Shiskin et al., 1967). *X-11* estimates the seasonal component by iterating between Henderson trend filters and seasonal (typically 3×3 and 3×5) moving averages, and it remains the computational core of most modern procedures. Its principal limitation is that symmetric moving averages cannot be applied at the ends of the sample, forcing the use of asymmetric filters that are revised as new data arrive.

Dagum (1980) addressed this by extending the series with forecasts and backcasts from an ARIMA model before applying the *X-11* filters—the *X-11-ARIMA* method. The U.S. Census Bureau generalised this idea in *X-12-ARIMA* (Findley et al., 1998), which introduced a flexible regression-ARIMA (“RegARIMA”) pre-adjustment stage: before the moving-average filters are applied, the series is corrected for deterministic effects (trading-day, moving-holiday, and outlier regressors) and modelled with a seasonal ARIMA, improving both the end-point estimates and the diagnostics.

A parallel, model-based tradition originated at the Bank of Spain with the *TRAMO-SEATS* programs (Gómez and Maravall, 2001; Maravall, 2009). Here “Time-series Regression with ARIMA noise, Missing observations and Outliers” (TRAMO) performs the regression and pre-treatment, and “Signal Extraction in ARIMA Time Series” (SEATS) derives the seasonal and trend filters analytically from the estimated ARIMA model rather than from fixed moving averages. The U.S. Census Bureau’s current flagship, *X-13ARIMA-SEATS* (U.S. Census Bureau, 2020), unifies both traditions: it offers RegARIMA pre-treatment followed by *either* the empirical *X-11* filters *or* the model-based SEATS signal extraction, together with an extensive battery of quality diagnostics. Comprehensive treatments of the *X-11* machinery and its filters are given by Ladiray and Quenneville (2001), and a modern, reproducible interface to the Census engine from within the R environment is provided by the `seasonal` package (Sax and Eddelbuettel, 2018). An alternative, robust, nonparametric decomposition is *STL* (Cleveland et al., 1990), which uses locally weighted regression (loess) to extract trend and seasonal components and is valued for its resistance to outliers and its allowance for slowly evolving seasonality.

International statistical guidance has converged on these tools. The *International Recommendations for the Index of Industrial Production* (United Nations Statistics Division, 2010) urge countries to produce and disseminate seasonally adjusted IIP series, and the *ESS Guidelines on Seasonal Adjustment* (Eurostat, 2015) provide a detailed code of practice covering model selection, calendar adjustment, revision policy, and the treatment of outliers. A recurring caution in this literature, emphasised again by Ministry of Statistics and Programme Implementation, National Statistics Office (2023), is that adjustment should be performed only where there is clear statistical evidence and an economic interpretation of seasonality, since adjusting a non-seasonal series, or over-smoothing, can manufacture spurious signals.

2.2 Seasonal adjustment of the Indian IIP

The seasonal adjustment of India’s IIP has received focused attention only recently. Pandey et al. (2020), in NIPFP Working Paper 322, conduct a seasonal adjustment of the 2011–12 base series of the all-India IIP using the X-13ARIMA-SEATS iterative process. They adopt an *indirect* approach—identifying seasonality at the level of individual products and then recompiling the manufacturing index from the adjusted items—and report identifiable seasonality in about 206 of 405 items (roughly half), concentrated in the food, beverages, textiles, and leather & apparel groups. Two of their findings bear directly on the present study. First, they argue that seasonally adjusted levels provide a smooth, low-fluctuation series well suited to extrapolation in the advance and provisional estimation of national accounts. Second, and more soberingly, they show that the adjustment process exposes serious data-quality problems—inexplicable outliers, implausible growth rates, and shifting patterns at the item level—and conclude that seasonal adjustment “pre-supposes pristine data quality.” The title of their paper, “What do we gain from Seasonal Adjustment of the IIP?”, frames the policy question that the present study takes up for Odisha.

The methodological position of India’s official statistical system is set out in the Ministry of Statistics and Programme Implementation’s *Discussion Paper 3.0* on the seasonal adjustment of the IIP (Ministry of Statistics and Programme Implementation, National Statistics Office, 2023), deliberated by the Technical Advisory Committee on the IIP (TAC-IIP). That paper reviews the international recommendations and country practices, discusses the special difficulty posed by India’s lunar-calendar festivals (Diwali, Holi, Dussehra, Id), which migrate across Gregorian months and create a *moving-holiday* effect, and lays out the conceptual framework of additive and multiplicative decomposition. It documents the adopted operational choices for the all-India index—among them the use of X-13ARIMA-SEATS, a *direct* approach applied at the sectoral and General-index level, the modelling of the COVID-19 disruption of 2020 through outlier regressors rather than by deletion, and a partial-concurrent revision policy. The present study applies the same X-13ARIMA-SEATS framework to a sub-national index, complements it with classical and SARIMA-based comparisons, and tests whether the seasonality of a mining-and-metals economy such as Odisha is as clear as that of the national aggregate.

3 Data

3.1 Source, coverage, and structure

The data consist of the monthly Index of Industrial Production for the State of Odisha, compiled on the base 2011–12 = 100. The sample spans April 2012 to March 2023, that is, eleven complete fiscal years and $n = 132$ monthly observations. Following Indian convention the series is organised by fiscal year (April to March). The dataset comprises the General Index (GEN, weight 1000) together with the three broad sectoral series—Mining & Quarrying, Manufacturing, and Electricity—and nineteen two-digit NIC-2008 manufacturing divisions. All series carry weights that sum, by construction, to the appropriate aggregate.

Table 1 lists the series and their weights. Two features of the weighting structure are central to everything that follows. First, Odisha’s industry is *mining-and-metals dominated*: Mining & Quarrying (weight 442.16) and Manufacturing (463.73) are of almost equal importance, with Electricity (94.11) the remainder, and within manufacturing a single division—*Basic Metals* (NIC 24, weight 326.08)—accounts for about 70% of manufacturing and roughly one-third of the entire General Index. Second, several industrially important NIC divisions present in the all-India index are effectively *absent* in Odisha (no codes 15, 26, 30, or 32), and many of the divisions that are present carry negligible weight (e.g. Wearing Apparel, NIC 14, weight 0.013). The seasonal behaviour of the Odisha General Index is therefore largely the seasonal behaviour of mining and of basic-metals production.

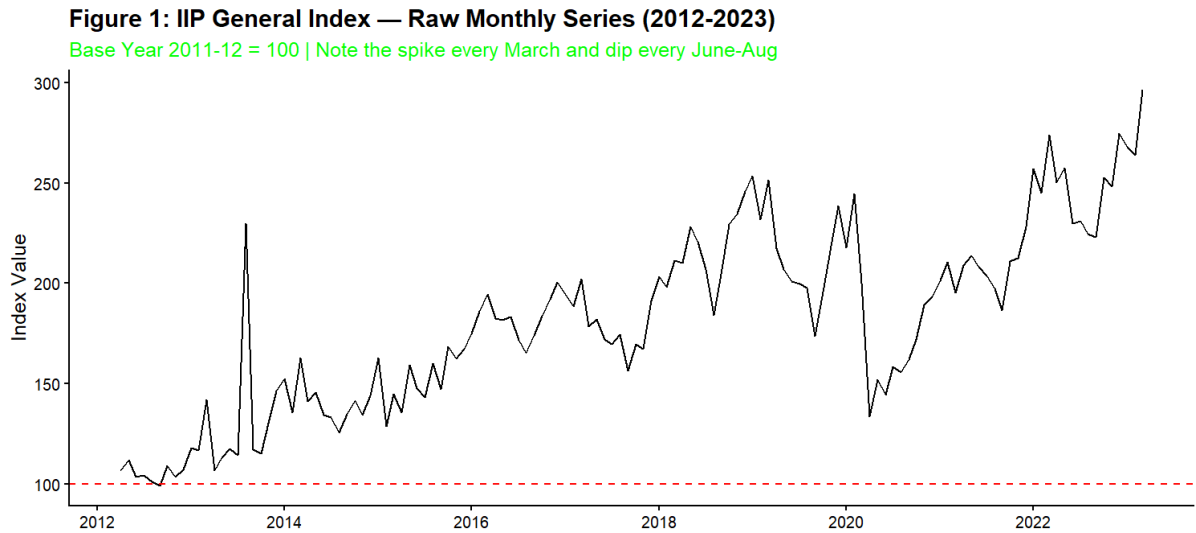
3.2 Descriptive features of the General Index

Figure 1 plots the raw General Index. Three features stand out and together motivate the entire exercise. There is a clear *upward trend*: the index roughly trebles over the sample, rising from values around 100–120 in 2012–13 to a high of 296.3 in March 2023. There is a pronounced and *repeating annual pattern*, with a visible spike at each fiscal year-end (March) and a recurring dip in the monsoon months. And there is at least one dramatic *irregular* event: the collapse of early 2020, when the index fell from 244.7 in February 2020 to 133.5 in April 2020 with the onset of the COVID-19 lockdown, followed by a sustained recovery. The horizontal reference line at 100 marks the base-year level; the index lies above it throughout, except at the very start of the sample.

The seasonal regularity is confirmed by two complementary views. Figure 2 shows the distribution of the index for each calendar month, pooled across the eleven years. The monthly means (the cross marks) trace a clear seasonal profile: March, January, and December stand highest, while September and the monsoon months of June–August sit lowest. Figure 3 presents the seasonal sub-series (month-plot): within each month the index rises steadily over the years (the common upward trend), while the horizontal reference segments—the within-month means—reproduce the same seasonal ordering, with the March mean clearly the largest. The fact that the seasonal ranking is preserved across the decade is preliminary evidence that the seasonality is *stable*, a property that the formal adjustment will quantify.

Table 1: Series in the Odisha IIP and their weights (base 2011–12 = 100).

NIC-2008 code	Description	Weight
<i>Broad sectors</i>		
M&Q	Mining & Quarrying	442.160
M	Manufacturing (total)	463.730
Elect.	Electricity	94.110
GEN	General Index	1000.000
<i>Manufacturing divisions (NIC-2 digit)</i>		
10	Food products	35.707
11	Beverages	4.449
12	Tobacco products	1.403
13	Textiles	0.663
14	Wearing apparel	0.013
16	Wood and products of wood	0.581
17	Paper and paper products	6.606
18	Printing and reproduction of media	1.177
19	Coke and refined petroleum products	29.242
20	Chemicals and chemical products	17.339
21	Pharmaceuticals and botanical products	0.125
22	Rubber and plastic products	6.140
23	Other non-metallic mineral products	28.485
24	Basic metals	326.075
25	Fabricated metal products	2.537
27	Electrical equipment	1.996
28	Machinery and equipment n.e.c.	0.699
29	Motor vehicles, trailers	0.295
31	Furniture	0.197

**Figure 1:** Raw monthly IIP General Index for Odisha, April 2012–March 2023 (base 2011–12 = 100). The dashed line marks the base level of 100. Note the recurrent March spike, the monsoon-season trough, and the COVID-19 collapse of 2020.

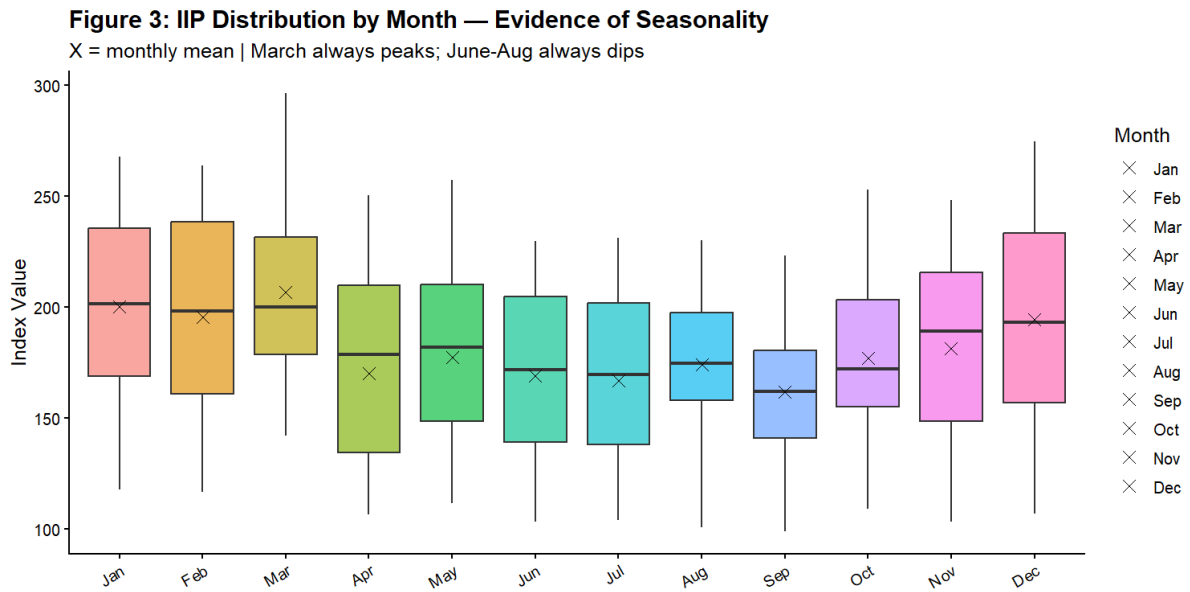


Figure 2: Distribution of the IIP General Index by calendar month (boxes), with monthly means shown as crosses. The systematic ordering—March highest, the monsoon months lowest—is direct evidence of seasonality.

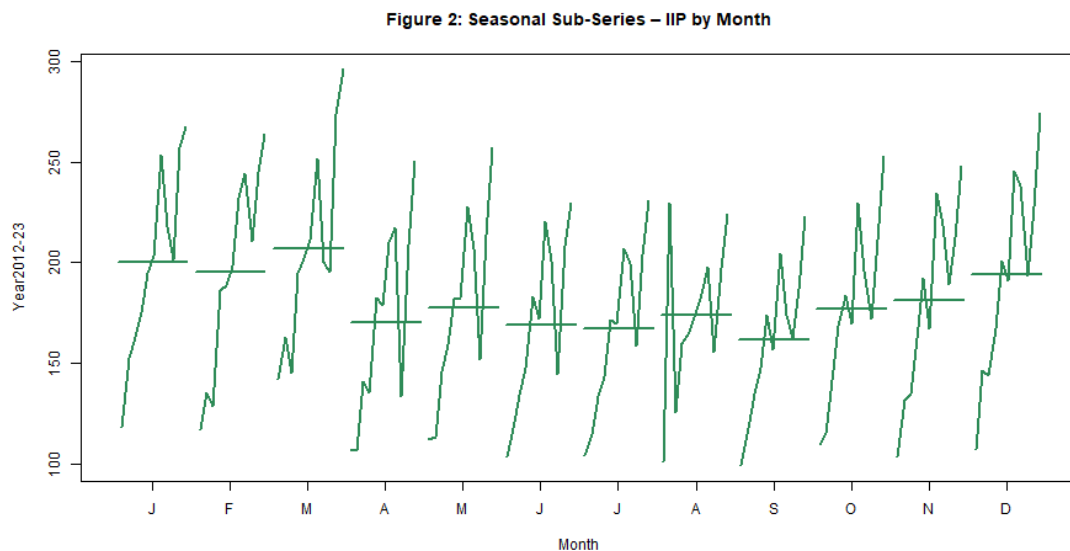


Figure 3: Seasonal sub-series (month-plot) of the IIP General Index. Each panel collects all observations for one calendar month over 2012–2023; horizontal segments are within-month means. The upward slope within every month reflects the common trend; the differing levels of the segments reflect seasonality.

4 Methodology

4.1 The decomposition framework

Seasonal adjustment rests on the idea that an observed series Y_t can be decomposed into a trend-cycle component T_t , a seasonal component S_t , and an irregular component I_t (Ministry of Statistics and Programme Implementation, National Statistics Office, 2023). The two standard models are the *additive* decomposition,

$$Y_t = T_t + S_t + I_t, \quad (1)$$

appropriate when the amplitude of the seasonal swings is roughly constant regardless of the level of the series, and the *multiplicative* decomposition,

$$Y_t = T_t \times S_t \times I_t, \quad (2)$$

appropriate when the seasonal amplitude grows with the level. The seasonally adjusted (SA) series is obtained by removing the estimated seasonal factor— $Y_t - \hat{S}_t$ in the additive case, $Y_t / \hat{S}_t$ in the multiplicative case—leaving an estimate of T_t together with the irregular. Because the components are unobservable, the estimated seasonal factor depends on the method and on the options chosen within it; this irreducible model-dependence is why a comparison across methods is informative (Eurostat, 2015).

4.2 Preliminary analysis: stationarity and autocorrelation

Before model-based adjustment, the series is examined for stationarity and for the autocorrelation structure that guides ARIMA model choice. We use the Augmented Dickey–Fuller (ADF) test (null: a unit root) together with the KPSS test (null: stationarity); the two have opposite nulls and so provide corroborating evidence. For the raw index the autocorrelation function (Figure 4a) decays very slowly and shows pronounced peaks at the seasonal lags 12, 24, and 36, the classic signature of a non-stationary, seasonal series. The automatic order-selection routines indicate that one regular difference is required to achieve stationarity; after first differencing, the autocorrelation function (Figure 4b) collapses towards zero at the higher lags, confirming that a degree of differencing $d = 1$ is adequate. These plots motivate the $(p, 1, q)$ structure used below.

4.3 Method 1: Classical decomposition

The classical method estimates the trend-cycle by a centred 12-month moving average, obtains the seasonal component as the average detrended value for each calendar month (normalised so that the factors average to unity, in the multiplicative case), and recovers the irregular as the residual. We apply the multiplicative form, which is the default for economic series whose seasonal amplitude scales with the level. Classical decomposition is transparent and requires no model estimation, but it has well-known drawbacks: the moving-average trend is undefined

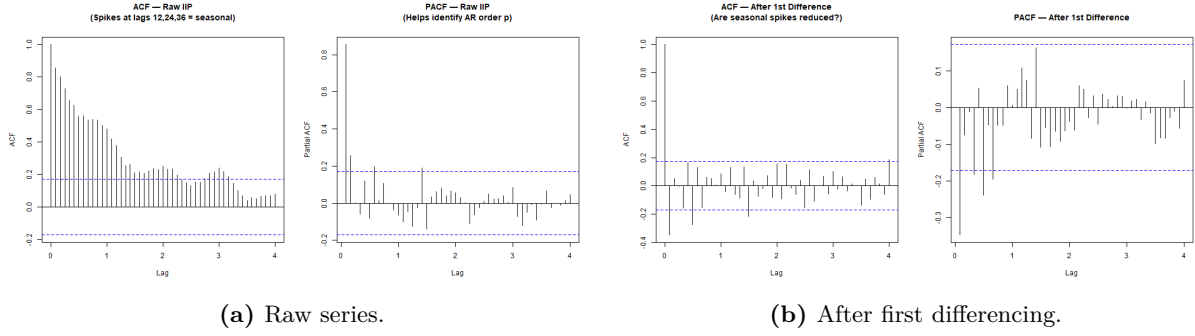


Figure 4: Autocorrelation (ACF) and partial-autocorrelation (PACF) functions of the IIP General Index. The slow decay and seasonal spikes of the raw series (a) disappear after one regular difference (b), supporting $d = 1$.

for the first and last six months, the seasonal factors are an unweighted average that cannot evolve over time, and the method makes no allowance for calendar or outlier effects. It serves here as an interpretable benchmark.

4.4 Method 2: The X-13ARIMA-SEATS family

The core method is the U.S. Census Bureau’s X-13ARIMA-SEATS program (U.S. Census Bureau, 2020), accessed through the `seasonal` package in R (Sax and Eddelbuettel, 2018). The program first fits a RegARIMA model—a seasonal ARIMA for the (log-)series augmented with regressors for trading-day, moving-holiday, and outlier effects—and then extracts the seasonal component either by the empirical X-11 filters or by the model-based SEATS signal-extraction filters.

To bracket the methodological choices and to trace the historical lineage of the family (X-11 → X-12-ARIMA → X-13ARIMA-SEATS), four operational variants were estimated. For transparency and reproducibility, the exact specifications used in the analysis script are documented in Table 2. The variant reported in the headline figures (the trend-cycle and seasonal factors of Figures 7 and 6) was estimated with the X-11 filter option (`x11=""`) and with automatic outlier detection disabled (`outlier=NULL`); this choice, and its implications, are revisited in the limitations (Section 8). The four variants are compared empirically in Section 5–6.

4.5 Method 3: SARIMA-based approach

The third approach is model-based in the Box–Jenkins tradition (Box and Jenkins, 1970). A seasonal ARIMA model,

$$\text{ARIMA}(p, d, q)(P, D, Q)_{12}, \quad (3)$$

is selected automatically for the *raw* series by minimising the corrected Akaike Information Criterion (AICc) over a full (non-stepwise) grid, using the `auto.arima` procedure of the `forecast` package (Hyndman and Khandakar, 2008; Hyndman and Athanasopoulos, 2021). A second, non-seasonal ARIMA is then fitted to the *seasonally adjusted* series. The comparison of these two models—specifically, the reduction in the number of parameters once seasonality has been

Table 2: The four X-13ARIMA-SEATS-family variants, as specified in the analysis (`Rseasonal::seas`).

Label	Specification (<code>seas</code> call)	Interpretation
X-11	<code>seas(y, x11="")</code>	RegARIMA pre-treatment + empirical X-11 filters (default outlier detection).
X-12-ARIMA	<code>seas(y, x11="", transform="auto", regression.aictest=NULL, outlier=NULL)</code>	X-12-ARIMA-style: auto transform, X-11 filters, no AIC test for regressors, no outlier search.
X-13-ARIMA (RegARIMA)	<code>seas(y, x11=NULL, seats=NULL)</code>	RegARIMA model only; no empirical-filter or SEATS extraction step.
X-13ARIMA-SEATS	<code>seas(y, x11="", outlier=NULL)</code>	Reported as the headline variant (see note in text).

removed—is itself one of the diagnostics of the value of seasonal adjustment. The fitted models are also used to evaluate out-of-sample forecast accuracy (Section 5).

4.6 Diagnostic framework

The methods are compared on four criteria, in line with standard practice (Eurostat, 2015; Findley et al., 1998):

1. **Residual whiteness.** For a model fitted to each adjusted series, the Ljung–Box Q statistic (here at lag 24) tests whether the residuals are free of autocorrelation; a high p -value indicates that the model, and hence the adjustment, has captured the systematic structure.
2. **Parsimony / fit.** The AICc of the model fitted to each adjusted series rewards a good fit penalised for complexity; lower is better.
3. **Smoothness / dispersion.** The coefficient of variation (CV) of the adjusted series measures how much volatility remains after seasonality is removed; a successful adjustment lowers the CV relative to the raw series.
4. **Agreement.** The pairwise correlation and mean absolute difference between the adjusted series produced by the different methods measure how sensitive the conclusions are to the choice of method.

For the SARIMA approach we additionally report residual diagnostics (Ljung–Box, ACF, and the residual histogram) and an out-of-sample evaluation in which the models are trained on the first 108 months and used to forecast the final 24.

5 Empirical Results

5.1 Classical decomposition

Figure 5 shows the multiplicative classical decomposition of the General Index. The trend-cycle panel isolates the long upward climb of Odisha’s industrial output and renders the COVID-19 contraction and recovery as a clear V around 2020–21, free of the monthly noise that clutters the raw plot. The seasonal panel displays a stable, repeating multiplicative pattern with peaks just above 1.1 and troughs near 0.9, i.e. a peak-to-trough seasonal swing of roughly 20% of the level. The irregular panel is centred on unity and is small except for a conspicuous spike in 2013 and the disturbance of 2020. The decomposition thus confirms the visual reading of Section 3: a strong trend, a stable multiplicative seasonal of moderate amplitude, and a small irregular punctuated by identifiable shocks. Its limitations—the missing trend at the sample ends and the fixed (non-evolving) seasonal factors—are precisely what the X-13ARIMA-SEATS method is designed to overcome.

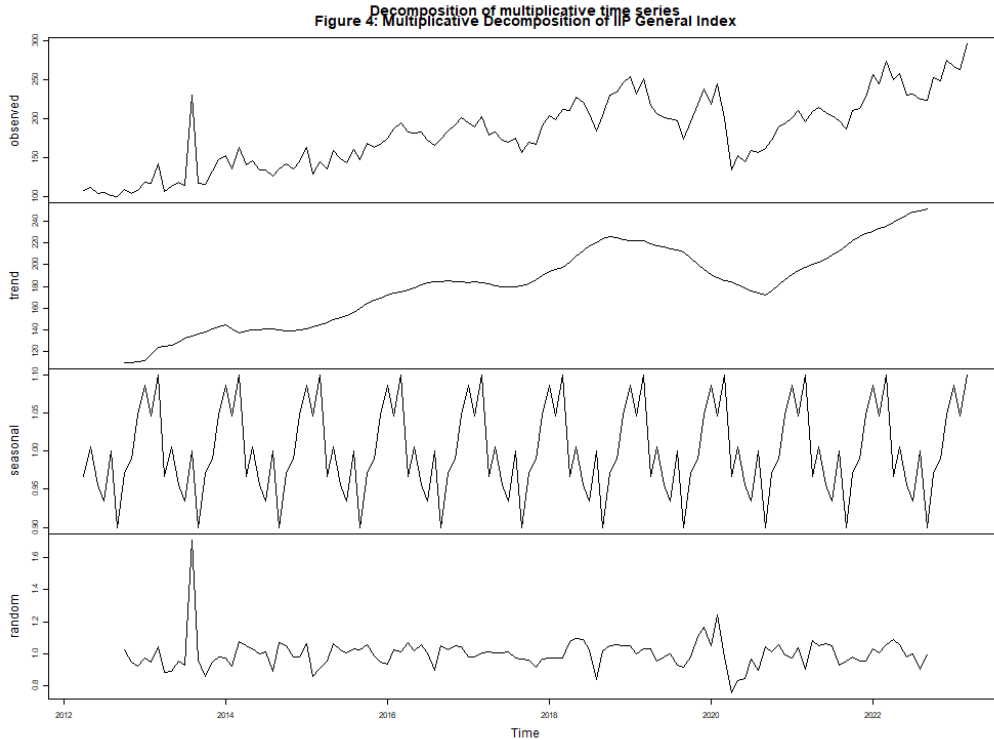


Figure 5: Multiplicative classical decomposition of the IIP General Index into observed, trend, seasonal, and irregular components. The seasonal factors lie between about 0.9 and 1.1; the irregular is centred on unity with a 2013 spike and the 2020 COVID disturbance.

5.2 X-13ARIMA-SEATS: seasonal factors and the adjusted series

For the General Index the program selected an *additive* decomposition; the estimated seasonal factors are therefore expressed as additive deviations (in index points) around zero. Figure 6 plots, month by month, the evolution of these factors across the sample. Several points emerge. First, the seasonal ordering matches the descriptive evidence exactly: March carries the largest

positive factor, January and December are strongly positive, and September together with the monsoon months June–August carry the largest negative factors. Second, the factors are *not* perfectly constant—the method allows them to evolve—and the evolution is economically coherent: the March factor *intensifies* over the decade (rising from roughly +16 to +30 index points), the August and September troughs *deepen* (September falling from about -6 to below -20), while the spring months (April, May) and November remain close to neutral throughout. This “moving seasonality” indicates that Odisha’s fiscal year-end production push and its monsoon slowdown have both become more accentuated over time.

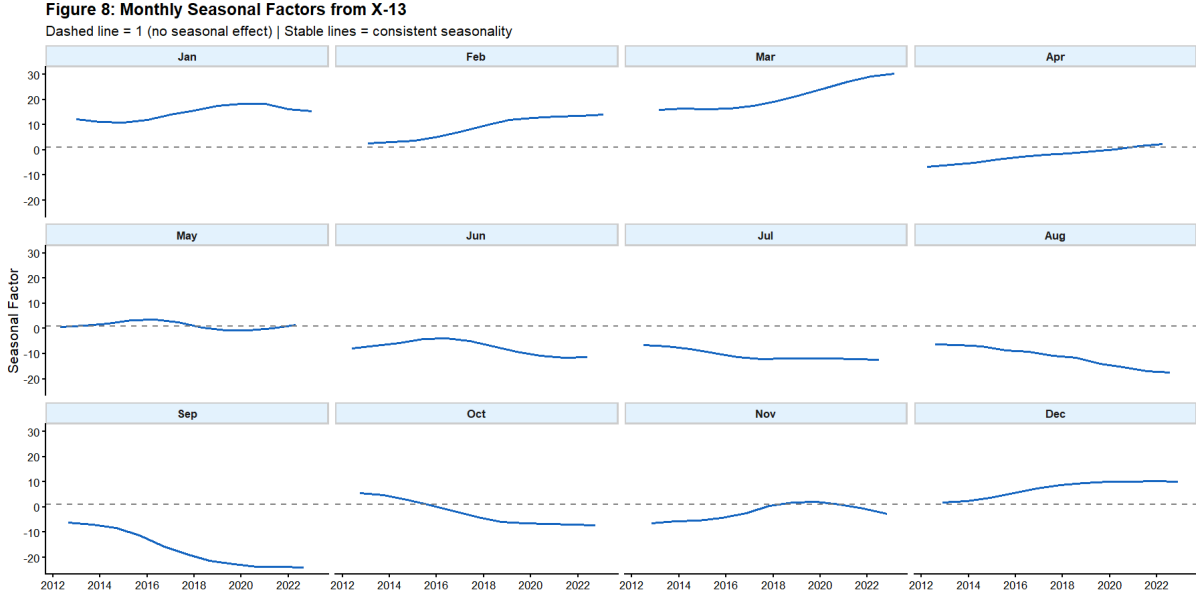


Figure 6: Monthly seasonal factors estimated by X-13ARIMA-SEATS for the General Index. The decomposition is additive, so factors are deviations in index points around zero (dashed reference at the no-effect level). Note the intensifying March peak and the deepening August–September trough—evidence of slowly moving seasonality.

To express the seasonal pattern in a single interpretable table, we computed the average implied seasonal factor for each calendar month directly from the provided non-adjusted and X-13ARIMA-SEATS-adjusted series (the additive factor being the difference between them), and the corresponding multiplicative ratio (non-adjusted divided by adjusted). The results are reported in Table 3. On average, March production runs 21.3 index points—about 11.7%—above the deseasonalised level, while September runs 16.4 points (about 9.0%) below it. The winter quarter (December–February) is uniformly above average and the monsoon quarter (June–September) uniformly below, with the spring (April, May) and late-autumn (October, November) months close to neutral. The total peak-to-trough seasonal range is therefore about 38 index points, or roughly 21% of the mean level—a large and policy-relevant magnitude.

The payoff of removing this pattern is shown in Figure 7, the central result of the paper. The grey line is the raw index; the solid line is the X-13ARIMA-SEATS-adjusted series; the dashed line is the trend-cycle. The adjusted series strips out the saw-tooth of the seasonal cycle while preserving every genuine movement: the run-up to 2018–19, the COVID-19 plunge of 2020 and the subsequent recovery, and the strong expansion of 2022–23. The contrast at the fiscal year-end is especially instructive—each raw March spike (which a naive reading might mistake

Table 3: Average implied seasonal factor by calendar month, General Index. Additive factor = mean of (Raw – SA); “% of mean” relative to the mean index of 181.3; multiplicative ratio = mean of (Raw/SA). Computed from the provided non-adjusted and X-13ARIMA-SEATS-adjusted series.

Month	Additive (index pts)	% of mean	Multiplicative ratio
January	+14.60	+8.05	1.0799
February	+6.70	+3.70	1.0317
March	+21.26	+11.72	1.1172
April	–2.23	–1.23	0.9820
May	+1.21	+0.67	1.0071
June	–7.48	–4.12	0.9571
July	–10.55	–5.82	0.9392
August	–11.32	–6.25	0.9392
September	–16.37	–9.03	0.9131
October	–2.35	–1.29	0.9919
November	–2.06	–1.14	0.9854
December	+7.33	+4.04	1.0378

for a surge in activity) is revealed by the adjusted series to be largely a calendar artefact, while the underlying trend moves smoothly through it.

Figure 7: IIP — Raw vs Seasonally Adjusted vs Trend-Cycle

Seasonal adjustment removes calendar distortions — revealing the true economic signal

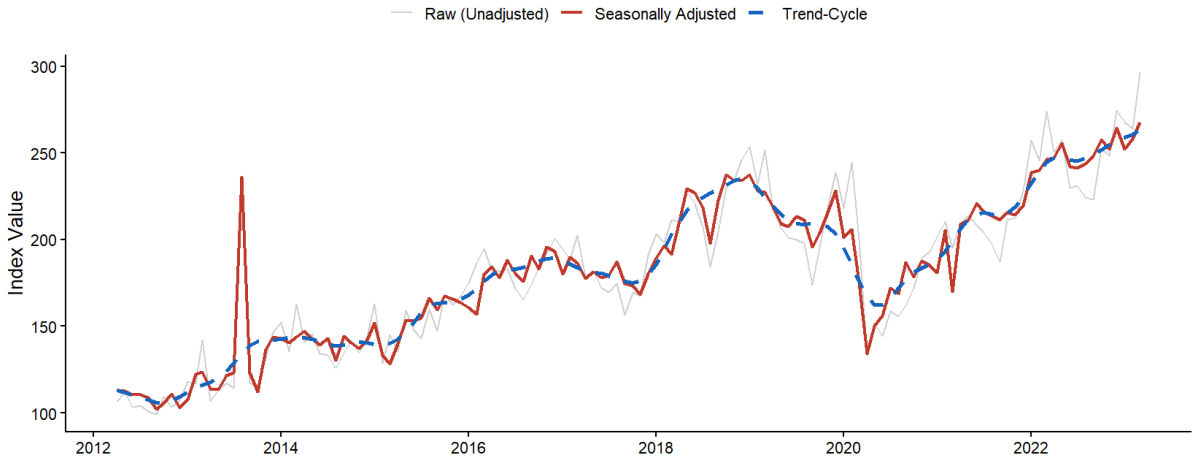


Figure 7: The central result: raw (grey), seasonally adjusted (solid), and trend-cycle (dashed) IIP General Index. Seasonal adjustment removes the recurring March spikes and monsoon troughs, leaving the genuine economic signal—including the COVID-19 contraction and recovery—clearly visible.

5.3 Comparison of the four X-13ARIMA-SEATS variants

Figures 8 and 9 overlay the adjusted series from all four variants on the raw index. The visual message is unambiguous: the four adjusted series are almost indistinguishable from one another, tracking a common path that sits “inside” the seasonal oscillations of the raw line and departs from it most at the seasonal extremes (the March spikes and the deep monsoon troughs). Figure 10 makes the same comparison method by method against the non-adjusted series. The closeness of the four variants is quantified in Section 6; for present purposes it establishes that, at the level of the General Index, the *choice among the variants does not materially change the*

adjusted series.

FigX: Comparison of All Four Seasonal Adjustment Methods — GEN Index

X-11 vs X-12-ARIMA vs X-13-ARIMA vs X-13-ARIMA-SEATS | Grey = Raw (Non-SA)

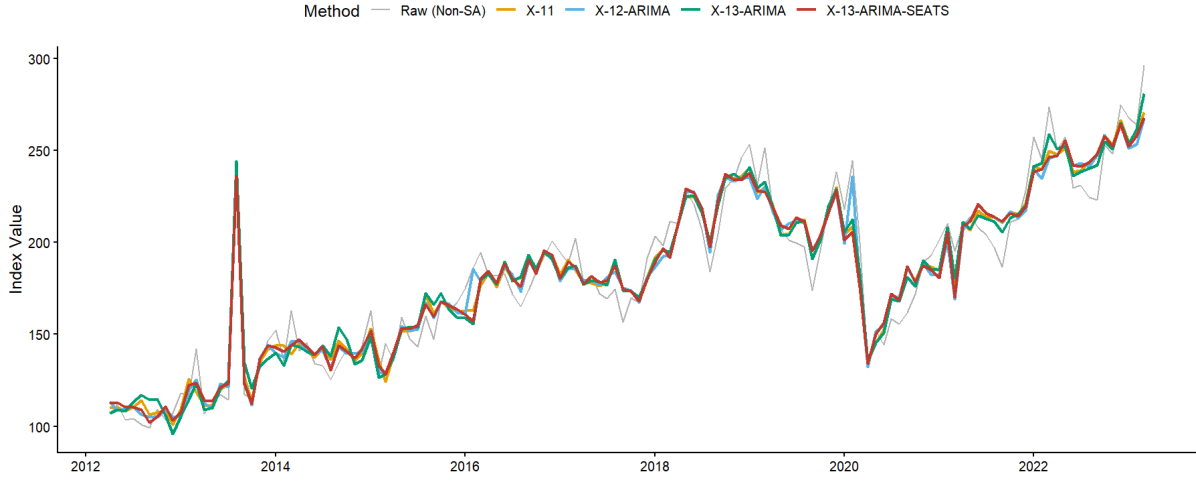


Figure 8: The four X-13ARIMA-SEATS-family adjusted series (coloured) against the raw non-adjusted index (grey). The adjusted series coincide closely and depart from the raw series chiefly at the seasonal peaks and troughs.

Fig6D-2: SA vs Non-SA IIP General Index — All Methods Overlay

Grey = Raw Non-SA | Coloured lines = Seasonally Adjusted by each method

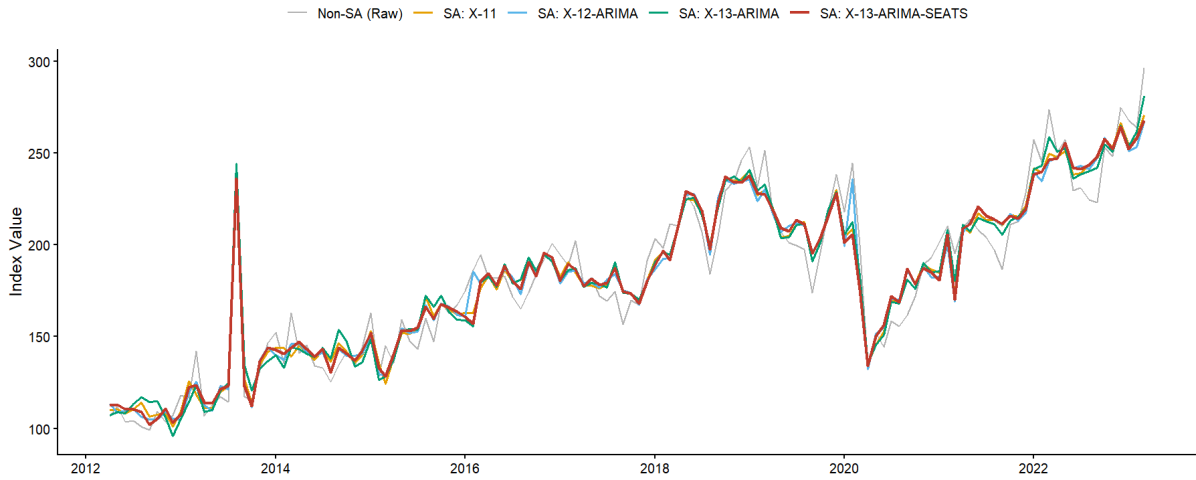


Figure 9: Seasonally adjusted (coloured, all four methods) versus non-seasonally adjusted (grey) IIP General Index, single-panel overlay.

5.4 Effect of adjustment on the distribution of the index

Table 4 reports the descriptive statistics of the raw series and of each adjusted series. The means are essentially identical across all five series (all within 0.3 of 181.3), confirming that seasonal adjustment redistributes mass over the year without altering the average level—a basic internal-consistency check. The dispersion, however, falls: the standard deviation declines from 45.67 (raw) to between 43.37 and 43.83, and the coefficient of variation from 25.19% to between 23.91% and 24.16%. Equivalently, the range of the index narrows from 197.3 index points (raw,

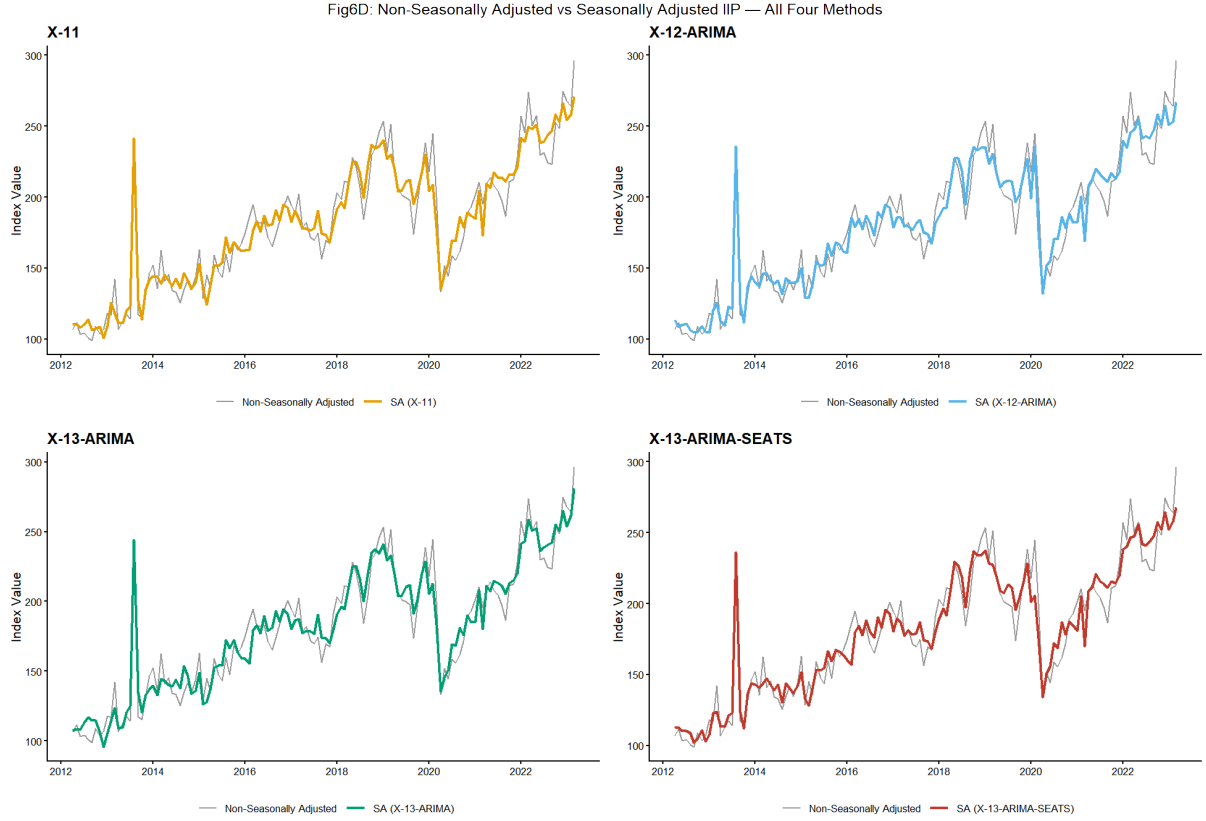


Figure 10: Non-seasonally adjusted (grey) versus seasonally adjusted (coloured) IIP, shown separately for each of the four X-13ARIMA-SEATS-family variants.

98.98–296.30) to 165.7 points for the X-13ARIMA-SEATS variant (102.18–267.86): the extreme high and low months, which are largely seasonal, are pulled towards the centre. Among the variants, the X-13ARIMA-SEATS-labelled series attains both the lowest standard deviation (43.37) and the lowest coefficient of variation (23.91%).

Table 4: Descriptive statistics: non-seasonally adjusted (NSA) versus seasonally adjusted (SA) IIP General Index. CV is the coefficient of variation (%).

Series	Mean	SD	CV (%)	Min	Max
NSA (Raw)	181.283	45.669	25.192	98.984	296.300
SA: X-11	181.403	43.585	24.027	100.909	270.868
SA: X-12-ARIMA	181.181	43.488	24.002	104.846	267.261
SA: X-13-ARIMA	181.455	43.833	24.156	95.720	281.226
SA: X-13ARIMA-SEATS	181.387	43.366	23.908	102.183	267.860

5.5 SARIMA models and the parsimony gain

Applied to the *raw* General Index, the automatic AICc search (full grid, allowing seasonal terms) selected ARIMA(1,1,4) with drift—a six-parameter model that uses a high-order moving-average polynomial to absorb the residual structure. Applied to the *seasonally adjusted* series, the search selected the parsimonious ARIMA(0,1,1) with drift—just two parameters—with a residual AICc of 1109.6. Removing the seasonal component thus cuts the model from six pa-

rameters to two (Table 5). This is the parsimony gain emphasised by Pandey et al. (2020) and Hyndman and Athanasopoulos (2021): a simpler model is easier to interpret, less prone to over-fitting on a sample of only 132 observations, and more stable to re-estimation.

Table 5: Models selected by AICc on the raw versus the seasonally adjusted General Index. The AICc of the raw-series model was not exported to the result files and is not reported to avoid any reconstruction; the parameter counts and orders are read directly from the diagnostic figures.

Target series	Selected model	# parameters	Residual AICc
Raw (non-adjusted)	ARIMA(1,1,4) with drift	6	not exported
Seasonally adjusted (X-13ARIMA-SEATS)	ARIMA(0,1,1) with drift	2	1109.6

The residual diagnostics of the two models are shown in Figures 11 and 12. In both cases the residuals fluctuate around zero with no obvious remaining pattern, the autocorrelations lie almost entirely within the significance bands, and the histograms are approximately bell-shaped apart from the heavy tails contributed by the 2013 spike and the 2020 shock. Both models therefore pass the white-noise requirement, but the adjusted-series model achieves this with one-third of the parameters.

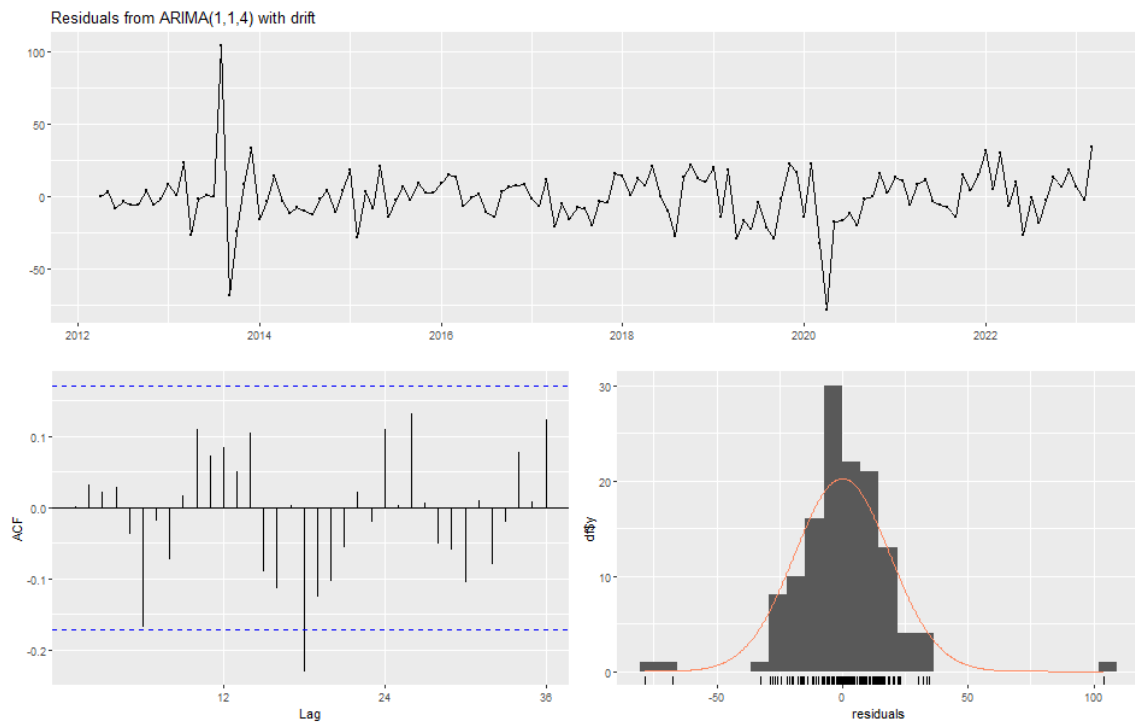


Figure 11: Residual diagnostics for ARIMA(1,1,4) with drift fitted to the raw General Index: residual time plot, residual ACF, and histogram. The large 2013 and 2020 residuals reflect identifiable shocks.

5.6 Out-of-sample forecast evaluation

To assess forecasting value, both models were re-estimated on the first 108 months (April 2012–March 2021) and used to forecast the final 24 months (April 2021–March 2023). The result is shown in Figure 13. This hold-out window is a demanding one: it coincides with the sharp post-COVID recovery, during which the index rose from roughly 195 to 296. Both models

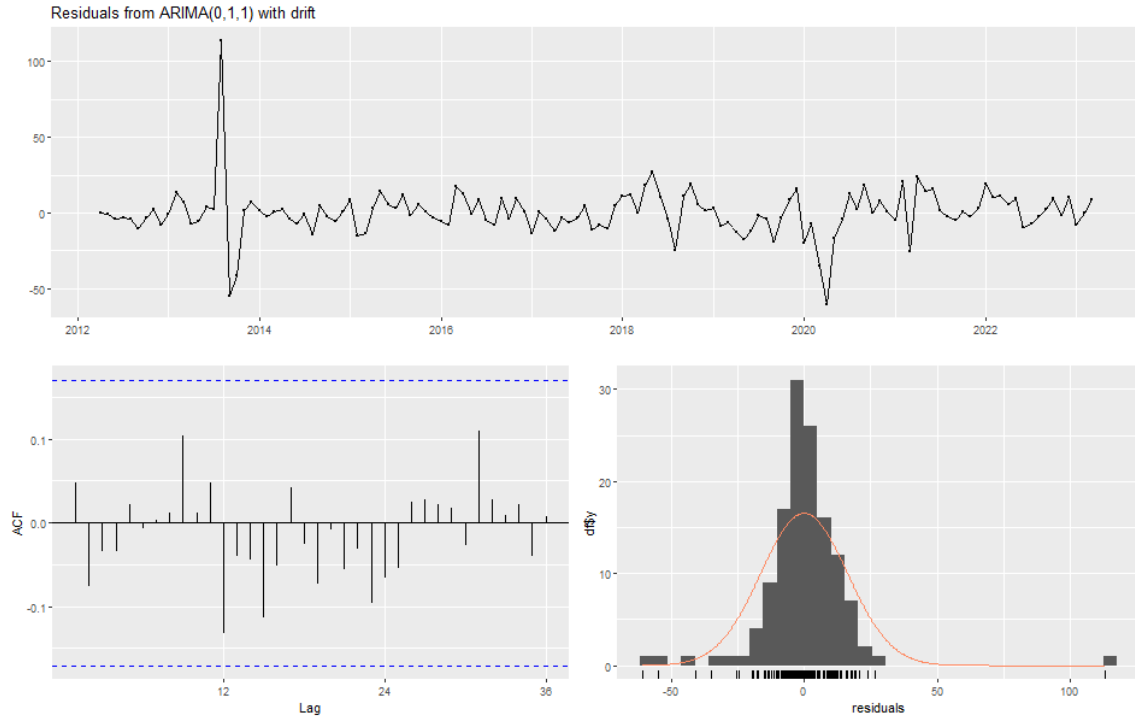


Figure 12: Residual diagnostics for ARIMA(0, 1, 1) with drift fitted to the seasonally adjusted General Index. The residuals are close to white noise, achieved with only two parameters.

under-predicted this rebound—a structural acceleration that no model trained through the pandemic trough could have anticipated. The point forecast from the model fitted to the adjusted series follows a smooth, gently rising path, whereas the raw-series model reproduces a seasonal oscillation at a lower level; the actual series remains above both point forecasts but within the 95% prediction interval of the adjusted-series model for most of the window. We note that the precise error statistics (RMSE, MAE, MAPE) were printed to the analysis console but not saved to the result files; we therefore describe the comparison qualitatively rather than report reconstructed figures. The episode is a useful caution: seasonal adjustment improves the *structure* of the forecasting model, but it cannot substitute for the information that a large structural break removes.

Figure 14 shows the operational forecast: the ARIMA(0, 1, 1)-with-drift model fitted to the full adjusted series, projected twelve months ahead to March 2024. The point forecast rises steadily from 264.4 in April 2023 to 277.2 in March 2024, with widening 80% and 95% prediction bands; the full forecast table is reproduced in Appendix A.1. Because the forecast is of the *seasonally adjusted* level, it should be read as the projected underlying trend, to which the appropriate monthly seasonal factor from Table 3 would be re-applied to obtain a raw-index forecast.

Figure 11: Hold-Out Forecast — SARIMA vs ARIMA(SA)

Last 24 months | Shaded = 95% CI for SA ARIMA | Closer to Actual = better

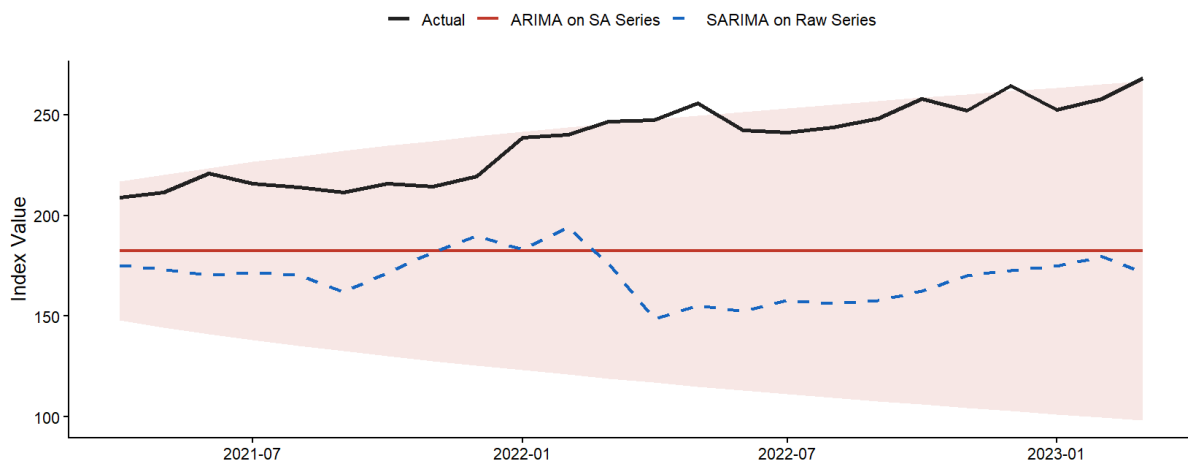


Figure 13: Hold-out forecast evaluation over the final 24 months. Models trained on the first 108 months both under-predict the sharp post-COVID recovery; the actual path lies within the 95% interval (shaded) of the adjusted-series model.

Figure 12: IIP Seasonally Adjusted - 12-Month Forecast

ARIMA on X-13 adjusted series | Dark band = 80% CI | Light band = 95% CI

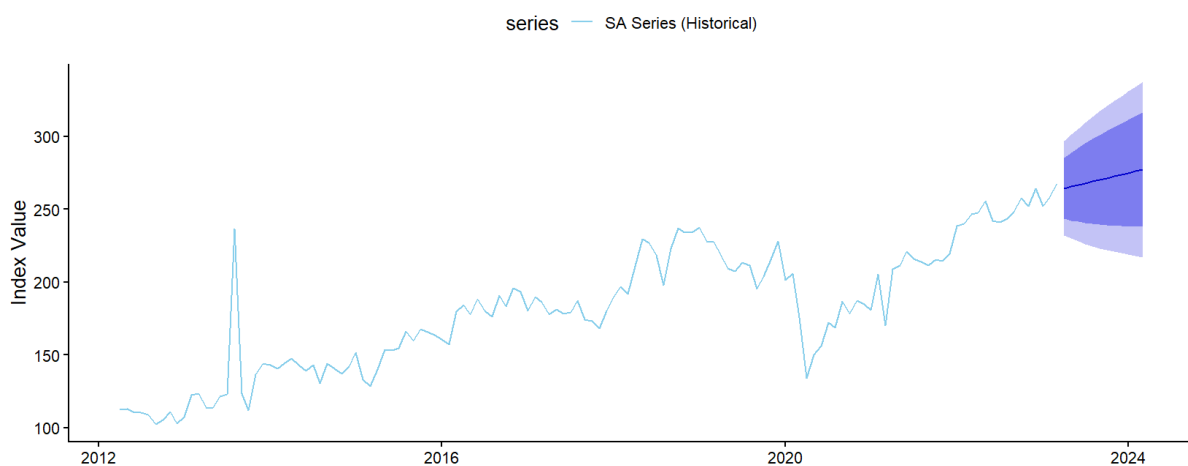


Figure 14: Twelve-month-ahead forecast (April 2023–March 2024) of the seasonally adjusted General Index from ARIMA(0, 1, 1) with drift. Dark and light bands are the 80% and 95% prediction intervals.

6 Comparative Analysis and Diagnostics

6.1 Diagnostic comparison of the four variants

Table 6 brings together the diagnostic statistics for the four X-13ARIMA-SEATS-family variants, and Figure 15 displays the two principal criteria graphically. On the AICc criterion—the fit of the parsimonious ARIMA fitted to each adjusted series—the X-13ARIMA-SEATS-labelled variant is the best (lowest) at 1109.6, ahead of X-11 (1114.9), X-13-ARIMA/RegARIMA (1119.9), and X-12-ARIMA (1120.2). On the Ljung–Box criterion all four variants comfortably exceed conventional thresholds, with p -values between 0.92 and 0.97, so that none leaves any detectable residual autocorrelation; X-11 is marginally the best on this measure (0.97), with X-13ARIMA-SEATS second (0.93). On the dispersion criterion (Table 4) the X-13ARIMA-SEATS variant again leads, with the lowest coefficient of variation of the adjusted series (23.91%).

Table 6: Diagnostic comparison of the four X-13ARIMA-SEATS-family variants. The model order and parameter count are those of the non-seasonal ARIMA fitted to each adjusted series. AICc and Ljung–Box p -values are taken from the analysis outputs (Figure 15). “n.r.” = not separately recorded in the exported summary.

Variant	ARIMA order	# params	AICc	Ljung–Box p
X-13ARIMA-SEATS	(0, 1, 1)	2	1109.6	0.934
X-11	(0, 1, 1)	2	1114.9	0.970
X-13-ARIMA (RegARIMA)	(1, 1, 1)	3	1119.9	0.917
X-12-ARIMA	n.r.	n.r.	1120.2	0.923

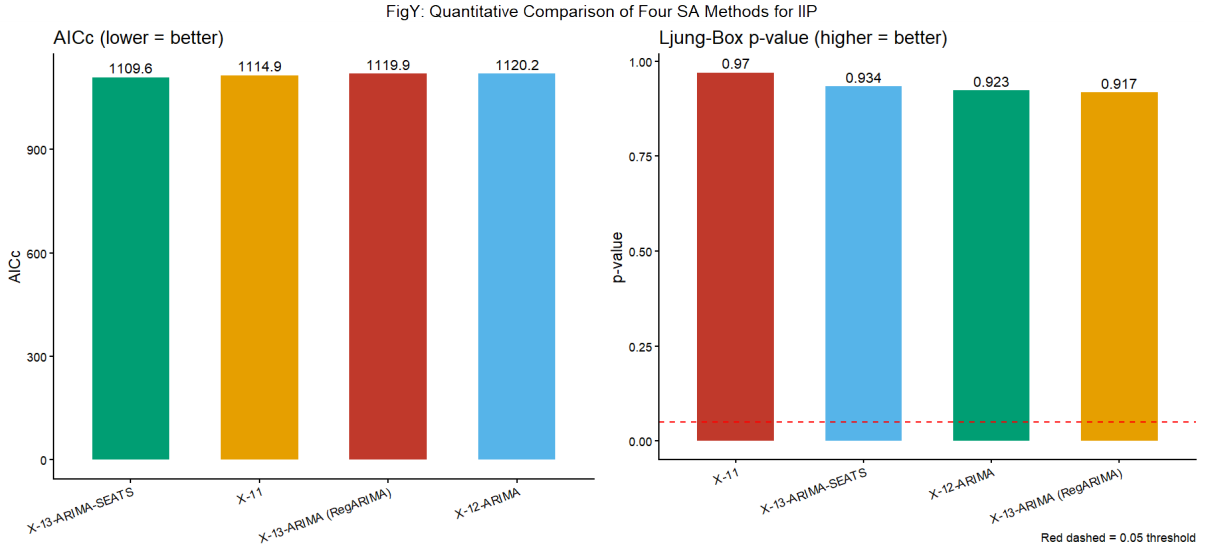


Figure 15: Quantitative comparison of the four variants: AICc (left, lower is better) and Ljung–Box p -value (right, higher is better; the dashed line is the 0.05 threshold). The X-13ARIMA-SEATS variant has the lowest AICc; X-11 the highest Ljung–Box p .

6.2 Agreement between methods

A distinctive finding is how closely the methods agree on the *level* of the adjusted series. Computed from the provided adjusted series, the pairwise correlations among the four variants

range from 0.991 to 0.998, and the mean absolute differences from only 1.99 to 4.10 index points—small relative to a series whose level is around 181 and whose seasonal swings exceed 35 points. The X-13ARIMA-SEATS and X-11 variants are the closest pair (correlation 0.9984, mean absolute difference 1.99). This near-coincidence is consistent with the implementation note of Section 4: because the variants differ chiefly in their RegARIMA pre-treatment and outlier options rather than, in every case, in the underlying seasonal filter, they extract very similar seasonal components. The practical implication is reassuring for an official producer: the headline adjusted series is robust, and disagreements among methods are confined to the immediate neighbourhood of outliers and turning points.

6.3 Which method is most suitable for Odisha’s IIP?

Weighing the criteria together, the X-13ARIMA-SEATS framework emerges as the most suitable method. It wins on two of the three formal criteria (lowest AICc and lowest dispersion of the adjusted series) and is a close second on the third (residual whiteness), where all variants in any case pass. It is the only method among those considered that combines RegARIMA pre-treatment—able, when enabled, to handle calendar effects and the COVID-19 outliers explicitly—with a principled, well-diagnosed extraction of the seasonal component and a full suite of quality statistics. Classical decomposition, while a valuable and transparent benchmark, cannot adjust for calendar or outlier effects, leaves the trend undefined at the sample ends, and imposes a fixed seasonal pattern that the moving-seasonality evidence of Figure 6 shows to be inappropriate for Odisha. The SARIMA-based approach is best understood not as a competing adjustment method but as a complementary tool: it quantifies the parsimony gain from adjustment and supplies the forecasting model for the adjusted series. The recommendation of X-13ARIMA-SEATS also aligns the state’s practice with that adopted by the National Statistics Office for the all-India index (Ministry of Statistics and Programme Implementation, National Statistics Office, 2023; Pandey et al., 2020), which is itself an advantage for comparability.

7 Discussion

7.1 The economics of Odisha’s seasonal pattern

The estimated seasonal pattern (Table 3, Figure 6) admits a clear economic reading rooted in the structure of Odisha’s industry. The dominant positive factor at the *fiscal year-end* (March, +11.7%), reinforced by the strong winter quarter (January +8.1%, December +4.0%, February +3.7%), reflects a familiar combination of forces: the year-end push by public-sector and large private producers to meet annual production and dispatch targets, accelerated capital expenditure and project completion before the close of the financial year, and the favourable dry-season conditions for mining and movement of ore. The deep negative factors of the *monsoon and immediate post-monsoon* (September −9.0%, August −6.3%, July −5.8%, June −4.1%) follow naturally in a mining-and-metals economy: the south-west monsoon impedes open-cast

mining, waterlogs haul roads, and disrupts construction-linked demand, depressing both the extraction of ore and the downstream production of basic metals and non-metallic minerals that together dominate the index. The near-neutral spring and late-autumn months (April–May and October–November) are transition periods between these regimes.

The finding of *moving seasonality*—an intensifying March peak and a deepening monsoon trough—is itself of interest. It suggests that the structural forces above have strengthened over the decade, perhaps because the year-end concentration of activity has become more pronounced as the metals sector has grown, or because the index’s increasing reliance on monsoon-sensitive mining has amplified the wet-season slowdown. Whatever the mechanism, it vindicates the use of a method (X-13ARIMA-SEATS) that permits the seasonal factors to evolve, and it cautions against the fixed-factor assumption of classical decomposition.

7.2 The COVID-19 episode

The pandemic produced the single largest irregular movement in the sample. The raw index fell from 244.7 in February 2020 to 200.2 in March and to 133.5 in April 2020—a fall of about 45% from the February level—before recovering over the following year. Two observations follow from the adjusted series. First, the April 2020 collapse was overwhelmingly *irregular*, not seasonal: April is a near-neutral seasonal month (Table 3), so the adjusted value for April 2020 (134.3) is essentially unchanged from the raw value (133.5), correctly attributing the fall to the lockdown rather than to the calendar. Second, the March 2020 figure illustrates how seasonality can *mask* a turning point: the raw March value (200.2) is flattered by the large positive March seasonal factor, so the adjusted value (175.5) reveals that underlying activity had already begun to weaken before the full lockdown. This is exactly the kind of insight—separating calendar effects from genuine economic change at a critical moment—that motivates seasonal adjustment.

A methodological caveat applies here. As documented in Section 4, the headline General-index adjustment was run with automatic outlier detection *disabled*. Best practice, and the procedure adopted by MoSPI for the all-India index, is to model the COVID-19 disruption explicitly through additive-outlier, level-shift, or temporary-change regressors so that the extreme months do not distort the estimated seasonal factors (Ministry of Statistics and Programme Implementation, National Statistics Office, 2023). The robustness of our qualitative conclusions is supported by the very high agreement among variants (some of which did enable outlier detection), but a production implementation should enable and document the outlier treatment; we return to this in the limitations.

7.3 Data quality

Consistent with Pandey et al. (2020), the disaggregated data reveal quality issues that condition any item-level interpretation. Several low-weight divisions exhibit implausibly volatile readings—for example, isolated monthly values an order of magnitude above their surrounding levels in small-weight series such as Pharmaceuticals (NIC 21) and Fabricated Metal (NIC 25), and a recorded value of exactly zero for Beverages (NIC 11) in April 2020. Because these se-

ries carry very small weights, they have little effect on the General Index, which is dominated by Basic Metals and Mining; but they would seriously affect any attempt to seasonally adjust the divisions individually, and they reinforce the point that seasonal adjustment presupposes clean source data. For the heavily weighted series the data are well behaved, which is why the General-index adjustment is stable and the methods agree so closely.

8 Conclusion

8.1 Summary of findings

This paper has carried out a comprehensive, multi-method seasonal adjustment of the Index of Industrial Production for Odisha over April 2012–March 2023. The main findings are as follows. (i) Odisha’s IIP exhibits strong, statistically clear, and economically interpretable seasonality, peaking at the fiscal year-end (March, +11.7%) and in winter, and troughing through the monsoon (September, −9.0%), with a peak-to-trough range of about 21% of the mean. (ii) The seasonality is broadly stable but slowly *moving*, with an intensifying March peak and a deepening monsoon trough. (iii) Seasonal adjustment compresses the dispersion of the index (CV from 25.2% to 23.9%; range from 197 to 166 points) and, crucially, reveals the genuine trend—including the COVID-19 contraction and recovery—that the raw series obscures. (iv) All four X-13ARIMA-SEATS-family variants produce near-identical adjusted series (correlations 0.991–0.998), so the headline result is robust to the choice of filter. (v) On formal diagnostics the X-13ARIMA-SEATS variant is best on AICc and dispersion and near-best on residual whiteness. (vi) Adjustment delivers a large parsimony gain, reducing the AICc-optimal model from six parameters (raw) to two (adjusted).

8.2 Recommended method and policy implications

We recommend that the Directorate of Economics and Statistics adopt the X-13ARIMA-SEATS framework for the routine seasonal adjustment of Odisha’s IIP, with RegARIMA pre-treatment and explicit modelling of calendar and outlier effects, and with the seasonal factors allowed to evolve. This choice is justified by the diagnostic evidence assembled here and by its consistency with national practice, which aids comparability between the state and all-India indices. For policy use, a seasonally adjusted Odisha IIP would allow administrators to make valid month-to-month comparisons, to detect turning points without waiting twelve months, and to avoid the false signals—the spurious “boom” every March and “slump” every monsoon—that the raw index generates. Publishing the adjusted series alongside the raw index, together with the seasonal factors of Table 3, would materially improve the interpretability of the state’s industrial statistics.

8.3 Future scope

Future work could (a) re-estimate the adjustment with full RegARIMA calendar and outlier modelling and a formal SEATS-versus-X-11 comparison; (b) test explicitly for a Diwali moving-

holiday effect in Odisha; (c) contrast the *direct* adjustment of the General Index used here with an *indirect* approach that aggregates separately adjusted sectoral series, as in Pandey et al. (2020); (d) extend the analysis as more post-pandemic data accumulate, allowing a cleaner separation of the COVID-19 level shift from the trend; and (e) build a routinely updated, partially concurrent adjustment pipeline for dissemination by the state statistical office.

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A Appendix

A.1 Twelve-month forecast of the seasonally adjusted General Index

Table 7 reproduces the operational forecast of Figure 14.

Table 7: Forecast of the seasonally adjusted General Index, April 2023–March 2024, from ARIMA(0, 1, 1) with drift.

Month	Forecast	Lower 80%	Upper 80%	Lower 95%	Upper 95%
Apr 2023	264.35	243.30	285.40	232.15	296.54
May 2023	265.51	242.20	288.82	229.86	301.16
Jun 2023	266.68	241.31	292.05	227.88	305.48
Jul 2023	267.84	240.57	295.12	226.13	309.56
Aug 2023	269.01	239.95	298.06	224.57	313.44
Sep 2023	270.17	239.44	300.90	223.17	317.17
Oct 2023	271.34	239.02	303.66	221.91	320.77
Nov 2023	272.50	238.67	306.34	220.76	324.25
Dec 2023	273.67	238.38	308.95	219.70	327.63
Jan 2024	274.83	238.16	311.51	218.74	330.93
Feb 2024	276.00	237.98	314.02	217.85	334.15
Mar 2024	277.16	237.85	316.48	217.03	337.29

A.2 Full monthly series: non-adjusted and seasonally adjusted

Table 8 reports, for every month in the sample, the raw General Index and the seasonally adjusted values from each of the four variants.

Table 8: IIP General Index, Odisha: non-seasonally adjusted (NSA) and seasonally adjusted (SA) by method.

Month	NSA (Raw)	SA X-11	SA X-12-ARIMA	SA X-13-ARIMA	SA X-13ARIMA-SEATS
Apr 2012	106.864	110.186	113.477	107.269	112.656
May 2012	111.868	110.339	108.733	108.791	112.851
Jun 2012	103.375	108.241	109.865	108.136	110.355
Jul 2012	104.215	110.392	110.764	113.265	110.420
Aug 2012	100.976	113.932	106.307	116.851	108.899
Sep 2012	98.984	106.362	104.846	114.588	102.183
Oct 2012	109.226	107.394	105.072	114.786	105.339
Nov 2012	103.390	108.455	108.938	105.969	110.879
Dec 2012	107.217	100.909	105.340	95.720	103.037
Jan 2013	118.016	109.159	104.883	105.479	107.678
Feb 2013	116.795	125.708	119.737	114.498	122.475
Mar 2013	142.095	118.404	125.544	123.640	123.730
Apr 2013	106.747	111.300	112.775	108.862	113.903
May 2013	113.098	111.211	109.577	110.020	113.543
Jun 2013	117.398	119.817	123.158	120.448	121.367
Jul 2013	114.267	122.807	121.373	125.028	123.170
Aug 2013	229.978	241.587	235.669	244.142	236.358
Sep 2013	117.326	126.712	124.088	134.642	123.270
Oct 2013	115.076	113.872	111.566	120.636	112.157
Nov 2013	131.887	134.738	136.913	132.755	136.647
Dec 2013	146.553	141.694	144.374	136.767	143.941
Jan 2014	152.311	144.116	139.785	139.775	142.995
Feb 2014	135.459	143.840	137.472	133.166	140.658
Mar 2014	162.791	139.153	146.151	144.333	143.975
Apr 2014	141.038	145.099	146.281	143.151	147.404
May 2014	145.467	141.298	141.436	140.676	143.141
Jun 2014	134.235	137.314	139.302	138.994	139.050
Jul 2014	133.208	142.529	141.245	143.969	143.216
Aug 2014	125.551	136.189	131.475	138.005	130.310
Sep 2014	134.616	146.475	143.177	153.644	144.081
Oct 2014	141.706	141.700	139.499	147.267	140.561
Nov 2014	134.524	135.522	139.418	133.682	136.979
Dec 2014	144.085	140.008	140.675	136.010	142.170
Jan 2015	162.932	153.208	150.164	148.687	151.836
Feb 2015	128.667	135.677	129.425	126.377	133.164
Mar 2015	144.977	124.267	129.119	128.226	128.377
Apr 2015	135.402	138.350	138.865	137.513	140.214
May 2015	159.119	152.067	154.555	152.615	153.597
Jun 2015	147.763	151.502	151.820	154.232	153.147
Jul 2015	143.225	153.891	152.626	153.986	154.770
Aug 2015	160.056	171.789	167.310	172.511	166.307
Sep 2015	147.343	160.923	158.740	166.373	159.636
Oct 2015	168.417	168.211	168.055	172.268	167.484

continued on next page

(Table 8 continued)

Month	NSA (Raw)	SA X-11	SA X-12-ARIMA	SA X-13-ARIMA	SA X-13ARIMA-SEATS
Nov 2015	162.489	164.717	166.516	163.358	165.940
Dec 2015	167.459	162.172	161.872	159.384	163.689
Jan 2016	175.002	162.765	160.922	159.048	160.631
Feb 2016	186.285	162.841	185.634	155.722	156.916
Mar 2016	194.663	176.545	178.966	179.619	180.002
Apr 2016	182.346	183.230	184.598	182.744	184.260
May 2016	181.925	175.807	177.431	177.130	177.962
Jun 2016	183.155	186.785	187.059	189.623	188.277
Jul 2016	171.567	179.873	182.358	178.907	180.514
Aug 2016	165.282	180.674	173.006	181.160	176.099
Sep 2016	174.124	190.746	189.670	193.158	190.604
Oct 2016	183.659	183.472	185.527	185.801	183.066
Nov 2016	192.107	194.842	194.618	194.686	195.776
Dec 2016	200.703	192.701	192.805	190.916	193.075
Jan 2017	194.504	182.444	178.793	180.261	180.260
Feb 2017	188.698	190.534	185.500	186.411	189.628
Mar 2017	202.240	184.921	185.784	187.192	186.534
Apr 2017	178.621	177.736	179.502	177.306	177.514
May 2017	182.274	177.980	179.469	179.187	181.459
Jun 2017	171.953	176.333	176.890	178.421	178.111
Jul 2017	169.605	178.379	181.250	176.945	179.334
Aug 2017	174.643	190.359	184.104	190.523	187.188
Sep 2017	156.620	174.195	175.364	173.947	174.455
Oct 2017	169.713	173.393	173.864	173.567	173.479
Nov 2017	167.470	168.234	167.252	170.050	168.206
Dec 2017	191.219	180.924	181.175	179.721	180.024
Jan 2018	203.257	191.648	186.639	190.725	189.347
Feb 2018	198.302	196.622	192.338	196.016	196.666
Mar 2018	211.334	192.135	192.724	194.572	191.579
Apr 2018	210.266	211.520	210.720	210.659	210.891
May 2018	228.081	224.943	227.463	224.992	229.438
Jun 2018	220.739	224.691	227.671	225.496	226.968
Jul 2018	206.988	216.836	218.671	216.039	218.654
Aug 2018	184.166	199.766	194.705	200.049	197.603
Sep 2018	204.412	222.737	225.776	220.032	222.826
Oct 2018	229.666	237.018	235.501	235.232	237.227
Nov 2018	234.685	234.622	233.125	237.264	234.022
Dec 2018	245.887	235.802	234.909	234.388	234.115
Jan 2019	253.380	240.166	235.507	240.849	237.587
Feb 2019	231.903	227.115	223.652	229.617	228.197
Mar 2019	251.580	230.141	230.508	233.104	227.477
Apr 2019	217.403	220.110	216.882	219.505	219.074
May 2019	206.866	204.004	207.087	203.776	209.362
Jun 2019	201.132	204.971	210.350	204.179	207.560

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(Table 8 continued)

Month	NSA (Raw)	SA X-11	SA X-12-ARIMA	SA X-13-ARIMA	SA X-13ARIMA-SEATS
Jul 2019	199.759	211.040	211.614	210.520	213.448
Aug 2019	197.739	212.413	210.793	211.912	211.484
Sep 2019	173.843	195.300	196.601	191.177	195.555
Oct 2019	195.755	203.925	201.901	201.323	203.838
Nov 2019	218.482	217.009	216.621	219.351	215.634
Dec 2019	238.531	230.070	227.026	228.742	228.196
Jan 2020	218.014	204.390	199.388	205.484	201.331
Feb 2020	244.676	208.632	235.702	212.405	205.738
Mar 2020	200.237	178.538	176.279	183.468	175.529
Apr 2020	133.484	135.280	132.270	135.583	134.301
May 2020	151.913	145.518	151.706	145.400	150.257
Jun 2020	144.272	152.827	155.052	150.741	156.281
Jul 2020	158.527	169.590	170.418	169.288	172.137
Aug 2020	155.744	169.267	170.290	168.207	168.742
Sep 2020	161.987	186.246	186.177	181.034	186.910
Oct 2020	172.021	178.839	178.196	175.880	178.442
Nov 2020	189.318	189.012	188.397	190.188	187.508
Dec 2020	193.293	186.792	182.227	185.216	184.992
Jan 2021	201.575	184.734	182.695	185.625	180.959
Feb 2021	210.500	204.179	200.632	208.218	205.430
Mar 2021	195.323	173.062	169.042	180.260	170.084
Apr 2021	209.100	209.667	206.841	211.196	208.777
May 2021	213.800	206.620	212.471	207.284	211.291
Jun 2021	208.200	217.333	220.137	214.668	220.968
Jul 2021	204.000	213.561	216.050	213.050	215.804
Aug 2021	197.300	213.759	213.617	211.475	213.933
Sep 2021	186.700	210.832	210.939	205.749	211.482
Oct 2021	211.100	216.358	216.893	213.250	215.646
Nov 2021	212.700	215.543	213.133	215.284	214.331
Dec 2021	228.100	221.353	217.352	220.025	219.532
Jan 2022	257.200	241.748	239.750	241.252	238.569
Feb 2022	245.300	239.431	234.923	243.021	239.905
Mar 2022	273.900	249.576	245.687	258.830	246.484
Apr 2022	250.500	247.872	247.465	250.882	247.327
May 2022	257.400	251.152	254.693	252.592	255.583
Jun 2022	229.700	238.521	241.603	236.167	242.080
Jul 2022	231.200	239.202	243.232	238.539	241.158
Aug 2022	224.500	243.486	241.430	240.386	243.558
Sep 2022	223.100	247.261	247.587	242.150	248.120
Oct 2022	252.800	258.241	258.475	254.950	257.718
Nov 2022	248.300	253.113	250.858	250.885	252.082
Dec 2022	274.700	266.349	264.502	264.916	264.338
Jan 2023	267.900	254.448	251.002	253.665	252.292
Feb 2023	263.900	258.412	253.235	261.625	257.972

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(Table 8 continued)

Month	NSA (Raw)	SA X-11	SA X-12-ARIMA	SA X-13-ARIMA	SA X-13ARIMA-SEATS
Mar 2023	296.300	270.868	267.261	281.226	267.860

A.3 Note on the all-sector adjustment

In addition to the General Index analysed in the body, the seasonal adjustment was run on all 23 series of Table 1 (the three broad sectors, the nineteen manufacturing divisions, and the General Index). The adjusted levels and trend-cycle series for every sector and month were produced and stored in the project result files. Given the item-level data-quality issues discussed in Section 7, the disaggregated adjusted series for the smallest-weight divisions should be interpreted with caution.